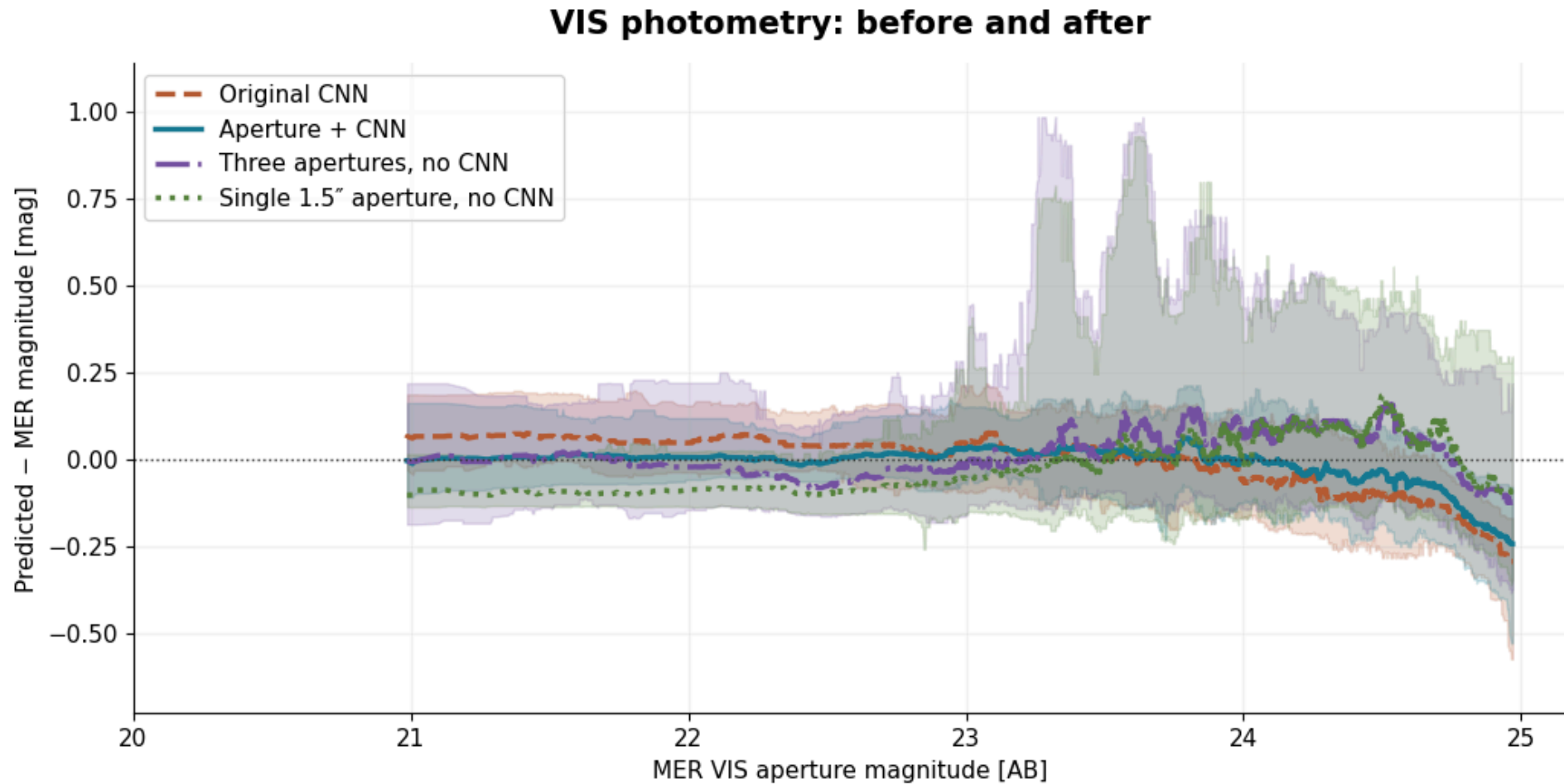


A simple CNN works

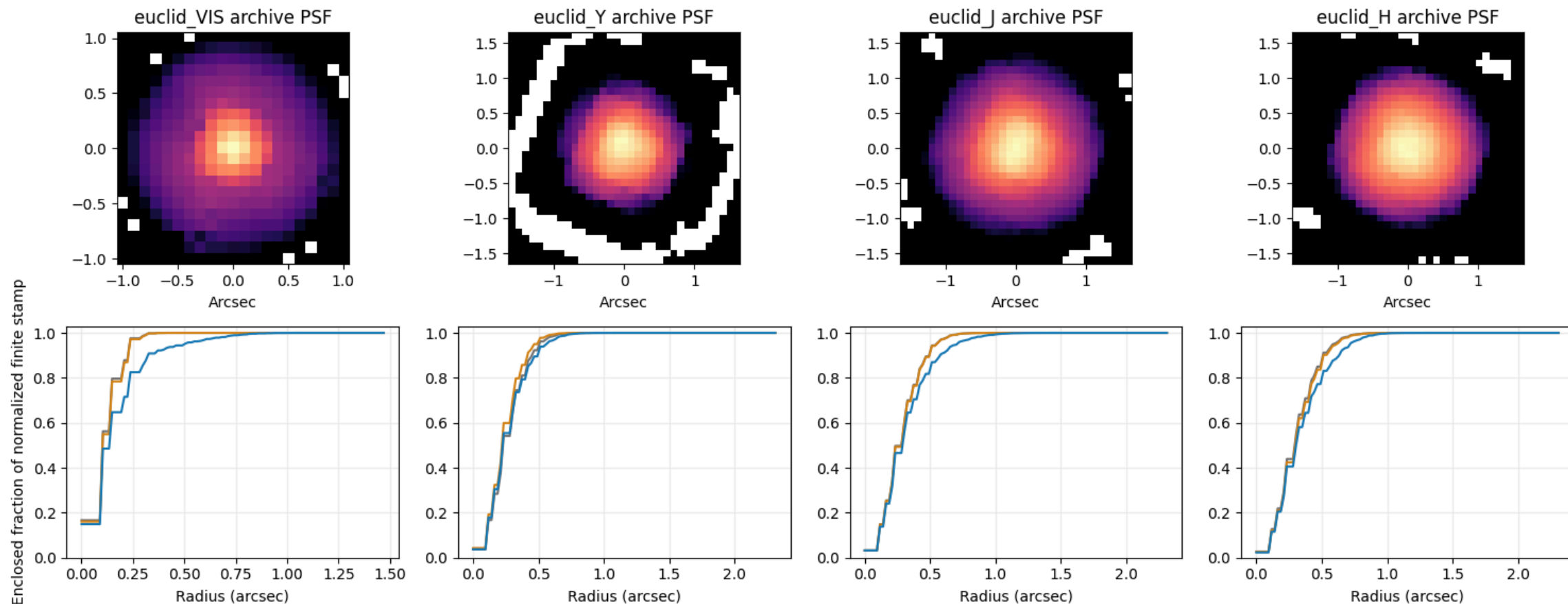


Aperture + CNN keeps VIS residuals close to MER across most of the tested range.

Use local PSFs

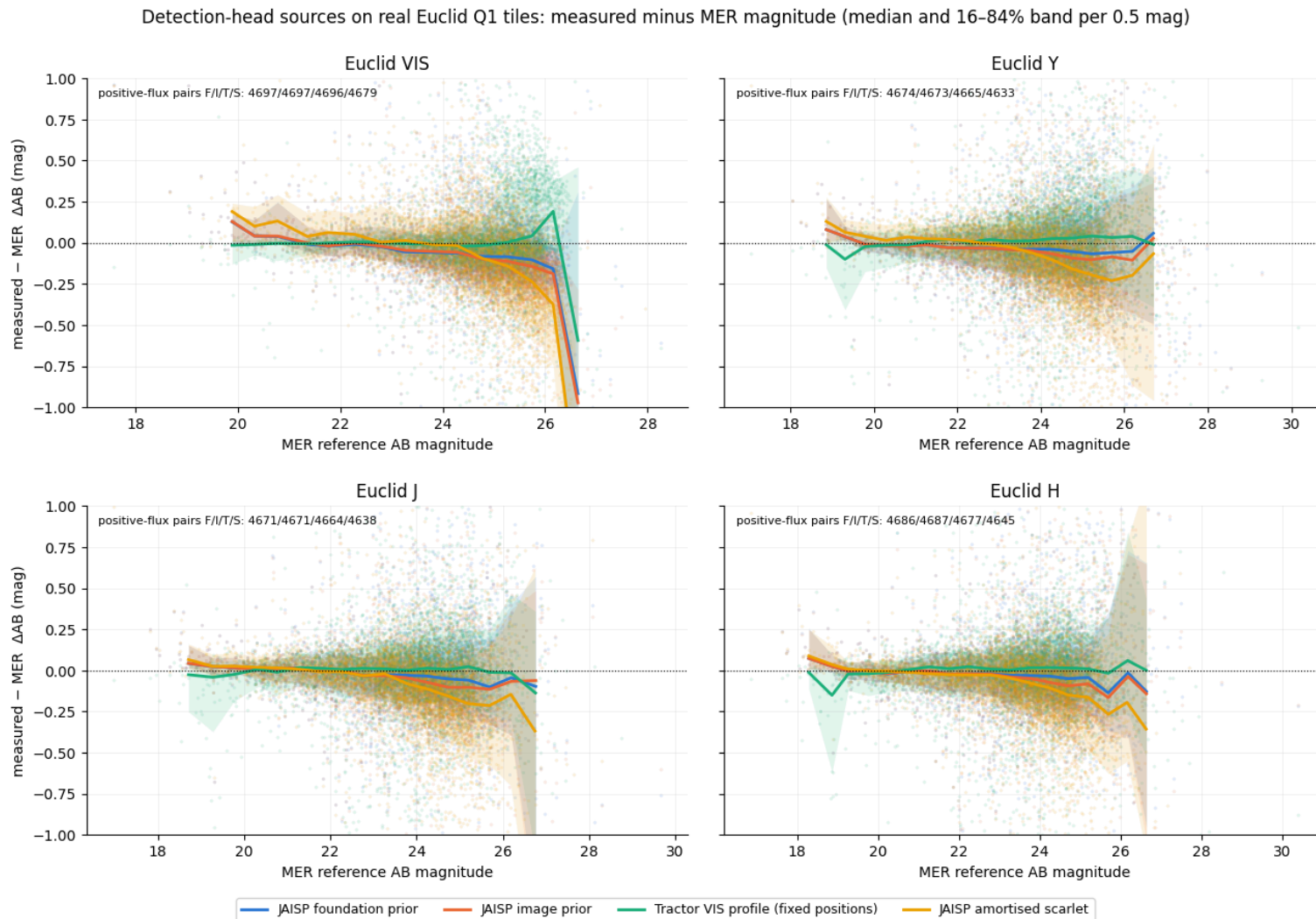
Euclid Q1 GRID-PSFs: example local samples and Gaussian approximations

— Original Gaussian — Local-FWHM Gaussian — Archive GRID-PSF



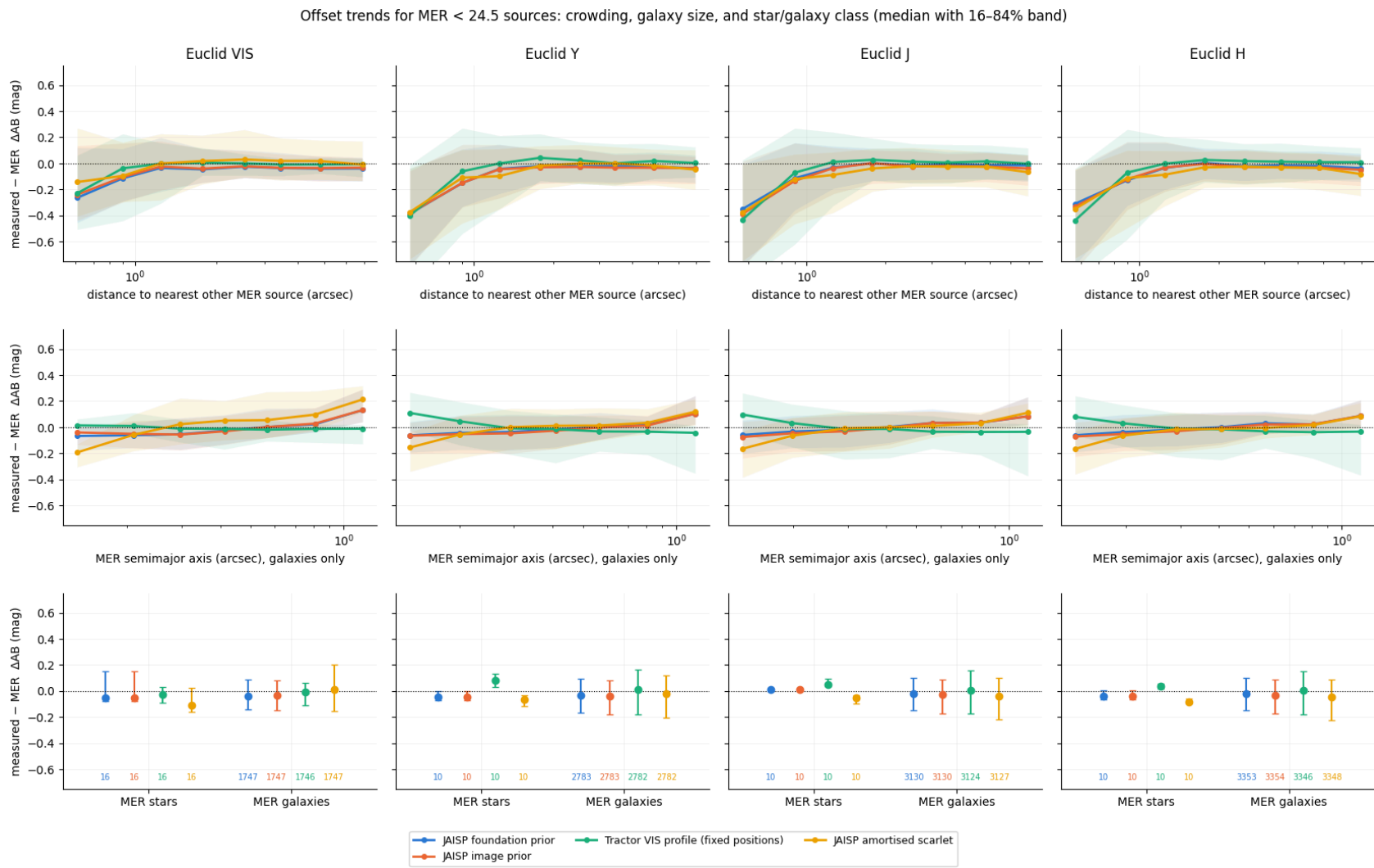
Euclid archive PSFs retain structure and extended wings beyond a Gaussian approximation.

Agreement with MER on real sources



Median offsets are small over much of the range; scatter and systematic shifts grow at the faint end.

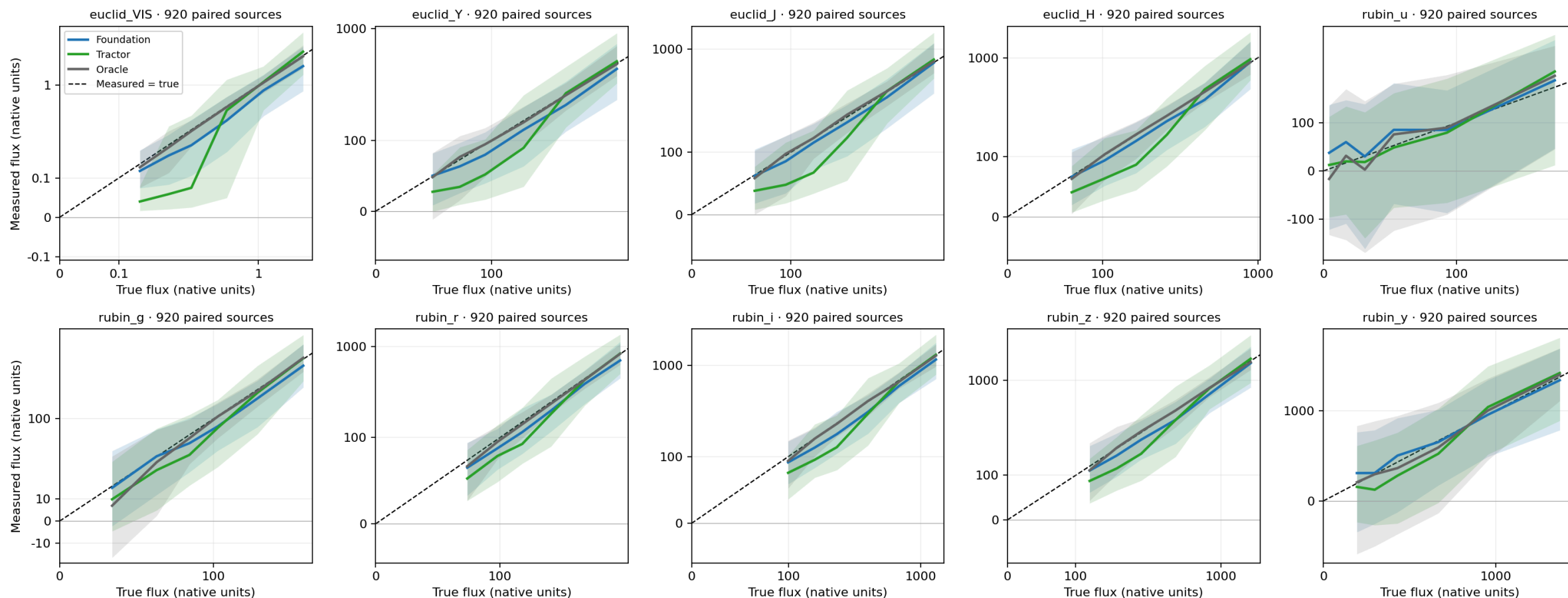
Crowding and galaxy size matter



Close neighbors produce the largest offsets; extended galaxies also show size-dependent trends.

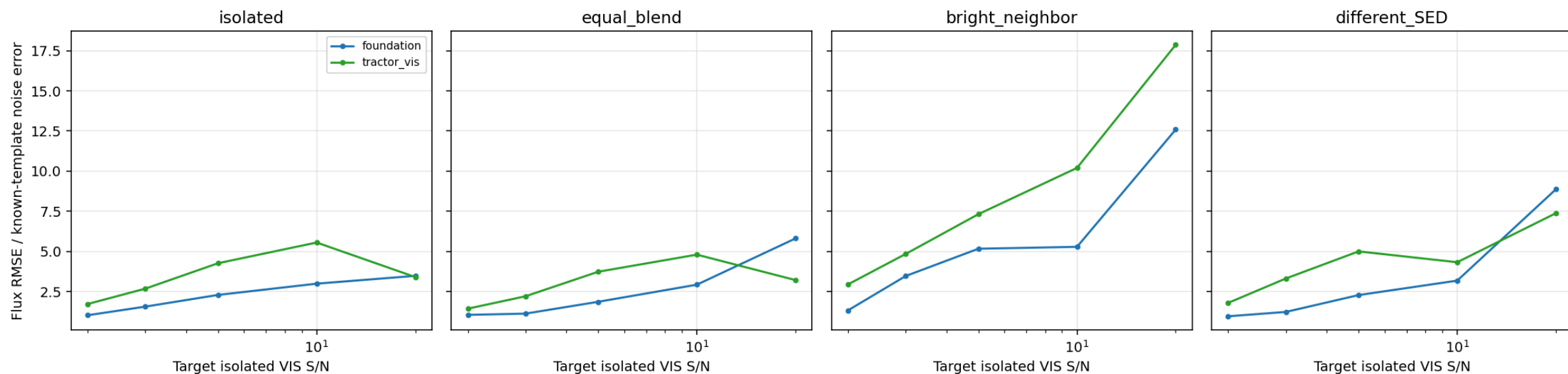
Known-flux injections test accuracy

Foundation, Tractor and oracle: median measured flux against injected truth
All qualified central sources · shading: 16th-84th percentiles



Foundation improves faint Euclid recovery over this Tractor baseline. Oracle fits the exact injected shapes.

Blending remains the hard case



Foundation has lower error in most faint bins; bright neighbors raise errors for both methods.